Ensemble Learning

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Ensemble Learning



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Learning Objectives

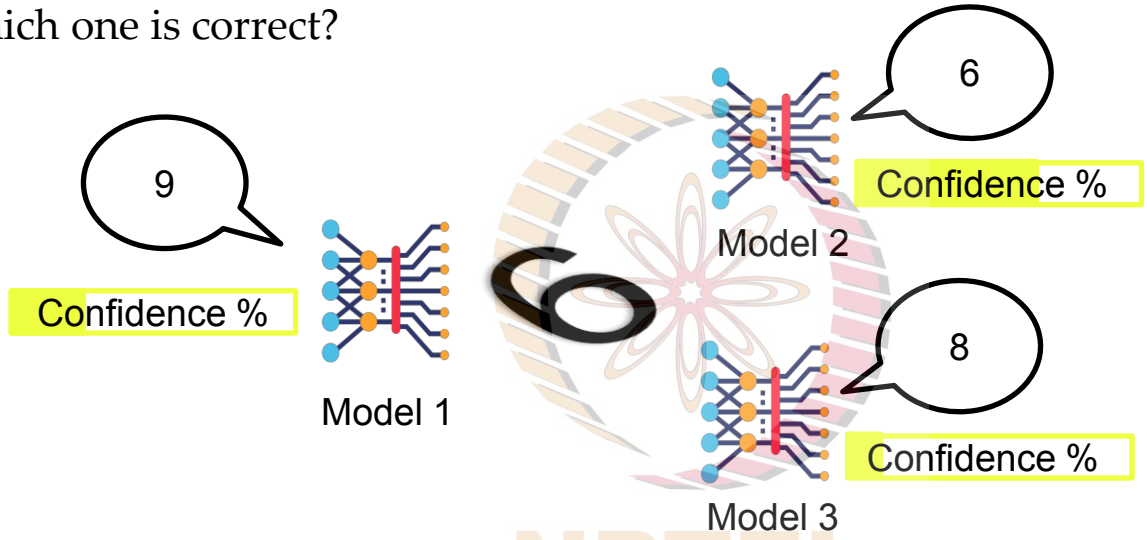
In this module, we will learn about Ensemble Learning:

- Motivation
- Introduction
- Issues with Single Classifiers
- Why Ensemble Learning works?
- Ensemble Learning Process
- Types – Homogeneous and Heterogeneous Methods
- Combining Strategies
- Ensemble Pruning

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Motivation

- Which one is correct?



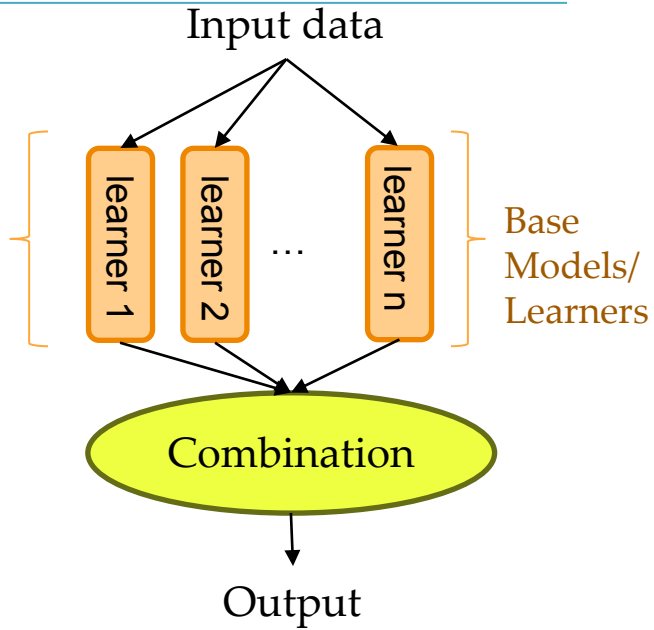
**The amount of green color in the bar shows the percentage of the decision confidence.*

- Let us combine multiple learners and have a poll ≡ Ensemble Learning

Ensemble Learning

- Involves combining multiple ML models (learners) together to create a stronger and more robust model that can handle a wider range of data complexity than any individual model
- Similar to **Committee-based learning**
- Based on the **principle of multiple explanations**

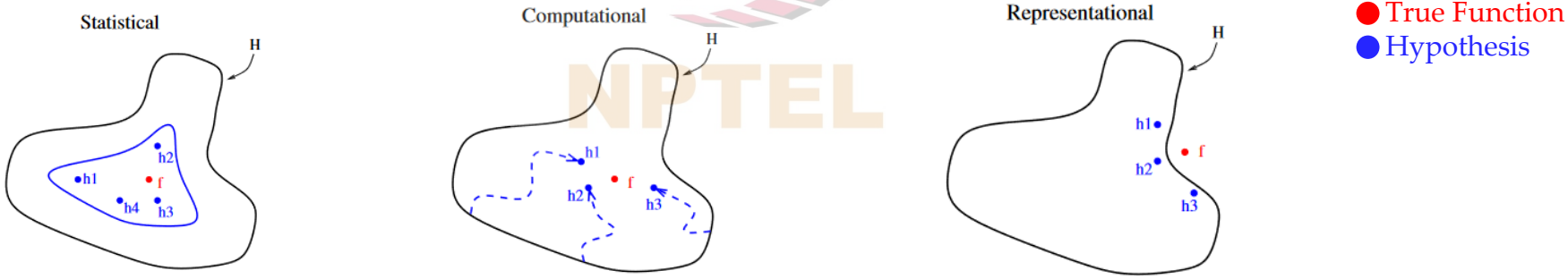
"If several theories are consistent with the observed data, retain them all."



Issues with Single Classifiers

- Three issues have been identified for which traditional learning approaches fail [Dietterich 2000]:

Statistical	Computational	Representational	
Hypothesis space is too large for limited training data	Algorithms that use local search may get stuck in local minima	True unknown hypothesis might not be represented by any hypothesis in the known hypothesis space	**Scenario**
• **Task**: Differentiate between lemons and oranges • **H1**: Lemons have a diameter of around 1.5" • **H2**: All oranges are orange in color • **Fail**: Some lemons are of big size; Some oranges are greenish in color	• **Task**: Differentiate between lemons and oranges • **H1**: All lemons are yellow in color • **H2**: All oranges are orange in color • **Fail**: Some lemons are green in color	• **Task**: Differentiate between lemons and oranges • **H1**: All lemons are sour • **H2**: All oranges are sweet • **Fail**: Some oranges are sour	**Example**



Source: Dietterich, Thomas G. "Ensemble methods in machine learning." *International workshop on multiple classifier systems*. Berlin, Heidelberg: Springer Berlin Heidelberg, 2000.

Issues with Single Classifiers (cont.)

Statistical	Computational	Representational	
• Several different hypotheses show similar performance on training data • Choosing one of these hypotheses brings the risk of selecting the wrong hypothesis for future data	Even if there is enough training data, selecting the best hypothesis may be difficult		**Consequence**
A learning algorithm that suffers from this issue is said to have a **High Variance**	A learning algorithm that suffers from this issue is said to have a **High Computational Variance**	A learning algorithm that suffers from this issue is said to have a **High Bias**	**Corollary**
Each hypothesis' votes can be averaged to reduce the risk of choosing the wrong classifier	By running the local search from various starting points, the combination may provide a better approximation to the true unknown hypothesis	Combining multiple hypotheses may expand the space of representable functions	**Towards Ensemble Learning**

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Why Ensemble Learning works?

- In general, there may not be a single learning algorithm that works best for all possible situations and hence, ensembling may work better.

- **Inductive Bias (IB)**: a set of assumptions a machine learning algorithm makes about the relationship between the input data and the output variable.
  - IB is used to guide the learning process by restricting the space of possible hypotheses
  - If the assumptions do not hold for the data, IB leads to errors
- Idea is that a better model might be suited for this data. This suggests combining multiple base learners.
- Combining several hypotheses reduces the risk of choosing the wrong local minima
- Combining several learning algorithms helps reduce model bias and variance

What Ensemble Learning can do?

- Can help to reduce overfitting
- Can provide more reliable predictions than individual models, as the combined model can account for variations and errors in individual models
- Can also help to identify and remove outliers and noise in the data, leading to more accurate predictions
- Often, many weak learners together beat the performance of single strong learner

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Ensemble Learning Process

- Ensemble learning is a two-step process:
  1. **Generating Diverse Learners**: Deals with generating multiple base learners that complement each other
  2. **Model Combination Schemes**: Deals with combining the output of base-learners for maximally improving evaluation metric

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Generating Diverse Learners

- **Aim**: To find a set of diverse learners who differ in their decisions so that they complement each other without compromising individual performances for the task
- Ways to achieve different learners:
  - **Different Learning Algorithms** (e.g., Logistic Regression and Decision Trees)
  - **Different Hyperparameters** for the same Learning Algorithm (e.g., Decision Trees with varying *max_depth*)
  - **Different Input Modalities** (e.g., only audio and only video analyzing models)
  - **Different Training Sets** (e.g., Bagging, Boosting, Cascading, etc.)

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Types of Ensemble Methods

- Based on the type of diversity in learners being combined, the ensemble methods are of two types.

Homogeneous Ensembles	Heterogeneous Ensembles
Build using multiple instances of the same base algorithm	Build using multiple instances of the different base algorithms

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Model Combination Schemes

Multi-Expert Combination	Multi-Stage Combination
Base-Learners work in parallel (Bagging)	Base-Learners work in series (Boosting)
Example: Mixture of Experts (MoE), and Voting	Example: Cascading
These are further of two types: 1. **Global Approach** [or **Learner Fusion**]: o Outputs generated by all learners are considered for the final decision o E.g., Voting & Averaging. 2. **Local Approach** [or **Learner Selection**]: o Outputs generated by one or a few learners are considered for the final decision o E.g., MoE (Using the Gating model)	The subsequent learner is trained with or tested on the instances where the current learner is not accurate/confident enough.

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Global Approach: Combining Strategies

- **Averaging**
  - Simple
  - Weighted

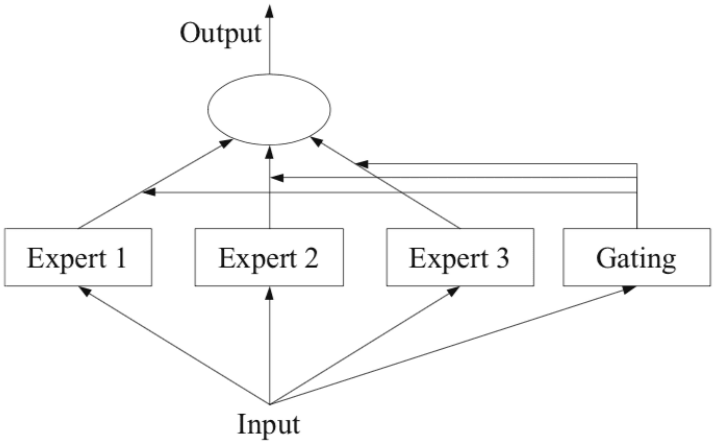
} Regression

- **Voting** ⟶ classification
  - **Plurality**: Class label receiving the largest number of votes becomes the final winner or majority vote
  - **Weighted**: More power to learners with higher performance
  - **Soft Voting**: Each learner produces a k-dimensional vector of class probabilities (k: Total number of classes)- decide based on the averaging of probabilities across learners or devise some other approach

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Local Approach: Mixture of Experts (MoE)

- Strategy: "Divide and Rule!"
- Combines multiple expert models to make predictions or decisions
- **Purpose**: To handle complex tasks by dividing them into sub-tasks, where each expert specializes in a particular aspect of the problem
- Three components:
  1. Each **Expert** is trained independently on a subset of the training data relevant to their specialization
  2. **Gating Network**: Learns to assign weights to each expert based on the input data to determine their contribution to the final prediction
  3. **Mixing Function** combines the predictions of the selected experts to generate the final output, e.g., Weighted Sum



Zhou, Zhi-Hua. *Ensemble methods: foundations and algorithms*. CRC press, 2012.

Local Approach: Mixture of Experts (MoE)

- Training:
  - **Initialization**: Experts and the gating network are initialized randomly or using pre-trained models
  - **Expectation-Maximization**: Alternates between updating the gating network and training the experts
- Advantages:
  - **Flexibility**: Allows the composition of different expert models, enabling better handling of complex tasks.
  - **Adaptability**: The gating network dynamically selects the most relevant experts based on the input, adapting to different problem contexts
  - **Improved Performance**: Can achieve better accuracy and robustness than individual experts

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Ensemble Pruning

- Strategy: "Many could be better than All!"
- Given a set of trained individual learners, rather than combining all of them, select a suitable subset for the ensemble
- Ensemble Pruning Techniques:
  1. **Weight-based Pruning**:
     - Removing models with low weights or contributions
     - Retaining models with higher importance
  2. **Error-based Pruning**:
     - Identifying models with high individual errors
     - Removing models that consistently underperform
  3. **Diversity-based Pruning**:
     - Analyzing models' diversity and redundancy of output
     - Eliminating models with similar behavior or predictions

Ensemble Pruning (cont.)

- **Advantages**:
  - Improved model generalization
  - Improves model interpretability
  - Reduces model complexity, computational resources, and memory requirements
  - Removes redundant or weak models to improve efficiency & performance

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To Summarize

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